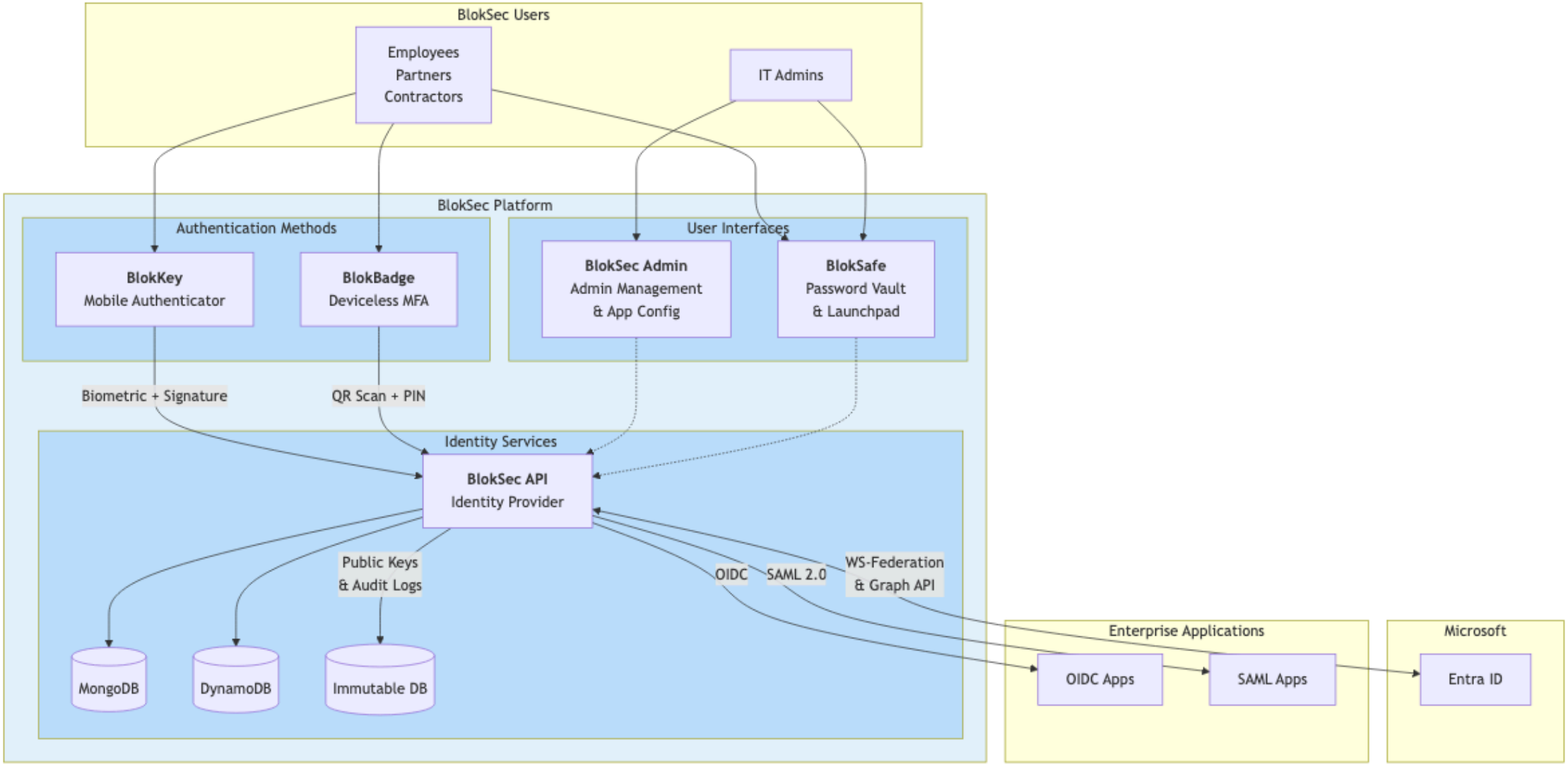
BlokSec

Passwordless Authentication for the Modern Enterprise

Phishing-resistant MFA for your entire workforce — from office to frontline

BlokSec Platform Architecture



Why BlokSec?

Passwordless Authentication for the Modern Enterprise

Eliminate Password Risk

- **No password database to breach** — Nothing to steal, nothing to crack
- **Phishing-resistant by design** — Cryptographic signatures can't be intercepted like codes
- **No shared secrets** — Unlike TOTP, secrets don't exist on both sides

Cover Your Entire Workforce

- **Office workers** get mobile-first BlokKey
- **Frontline workers** get deviceless BlokBadge
- One platform, every employee

Reduce IT Burden

- **Zero password resets** — The #1 help desk call, eliminated
- **No TOTP troubleshooting** — No clock sync issues, no lost authenticator apps
- **Self-service PIN rotation** — Users change PINs without IT involvement

Comply with Confidence

- **Immutable audit logs** — Tamper-proof records for regulators
- **Forensic integrity** — Logs can't be altered after the fact
- **Works with existing apps** — Standard protocols, no custom integration

Two Methods for Every Workforce

BlokKey

Mobile-First Authentication

- Smartphone app with biometric unlock
- Keys stored in TPM/Secure Enclave
- HD wallet for per-app key isolation

3 Authentication Factors

1. **Possession** — Device
2. **Biometry** — Face/Fingerprint
3. **Intent** — User action

Best for: Office workers, knowledge workers

BlokBadge

Deviceless MFA

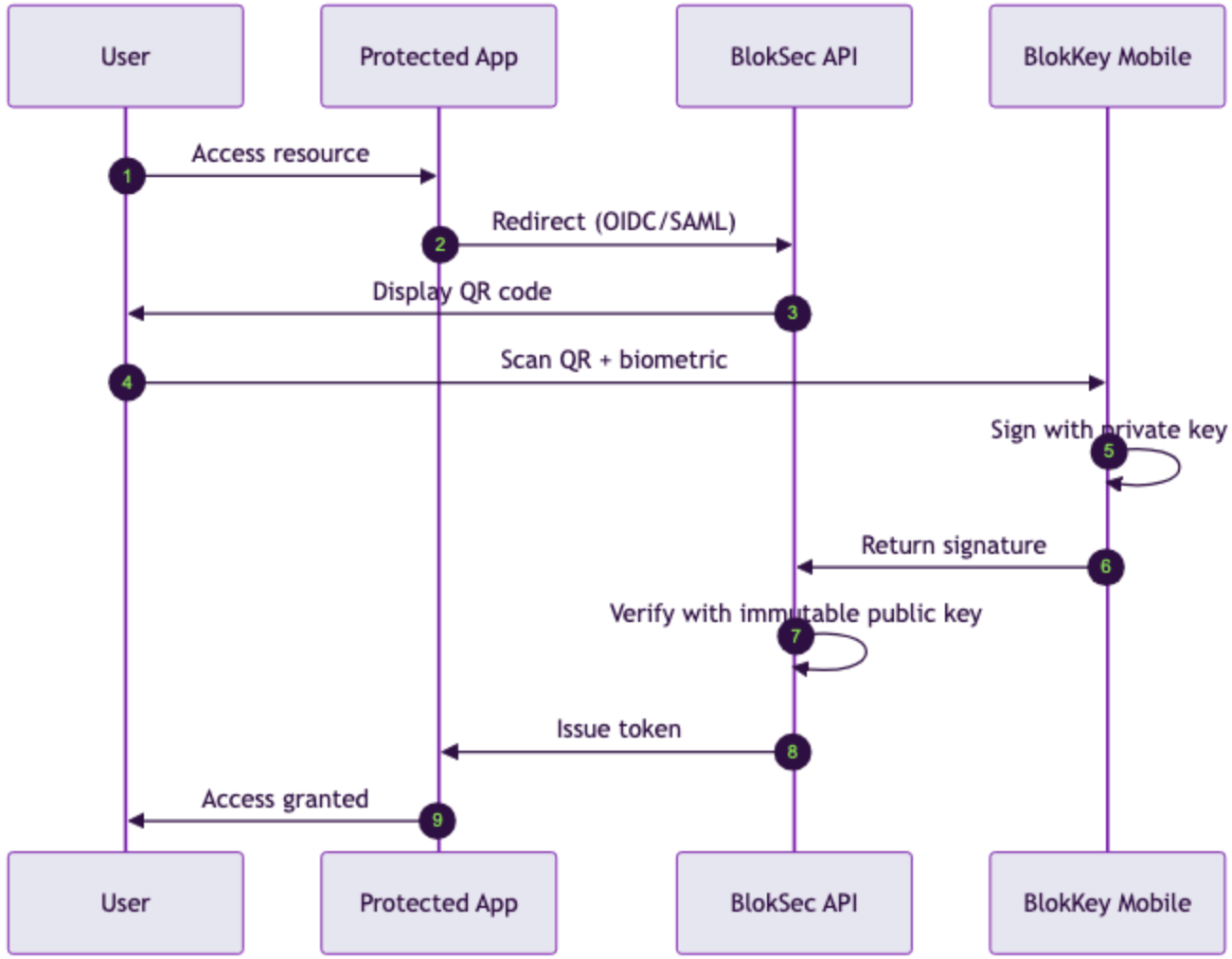
- Printed QR badge + PIN/passphrase
- No smartphone required
- Change PIN without reprinting badge

2 Authentication Factors

1. **Possession** — Badge
2. **Knowledge** — PIN/Passphrase

Best for: Frontline, retail, healthcare, manufacturing

BlokKey: Mobile-First Passwordless



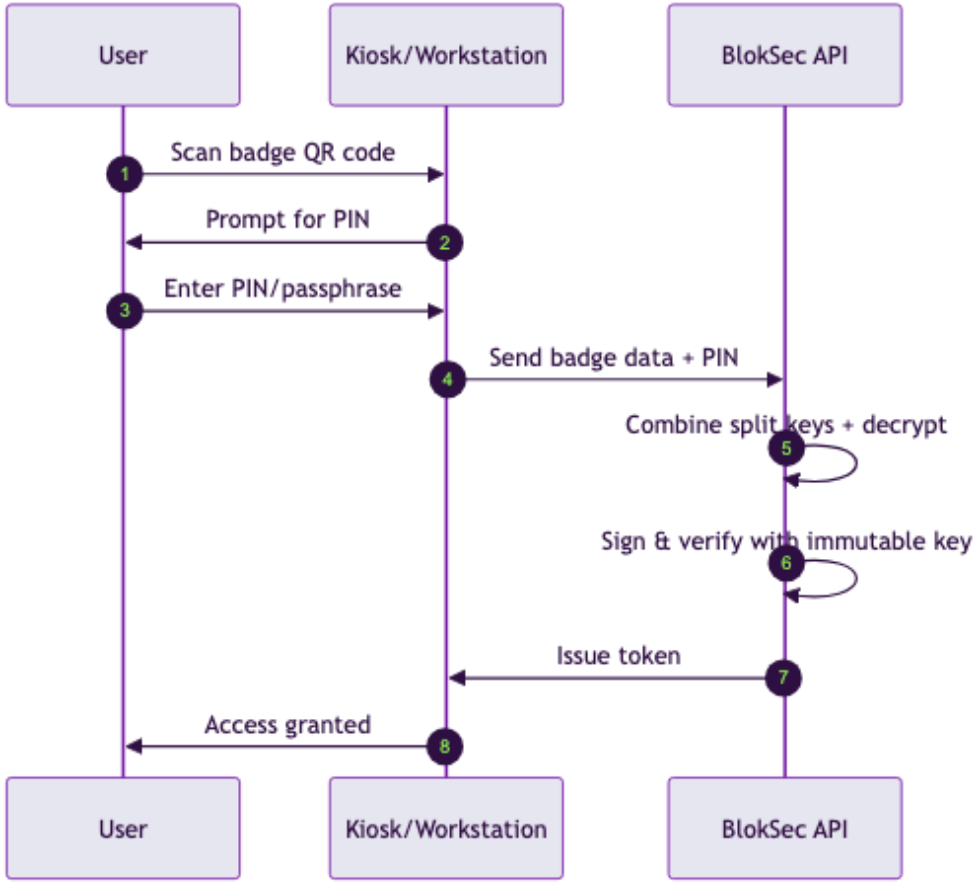
Why it matters

Private keys never leave the device.

No server-side secrets to breach.

Biometric unlocks signing — attackers can't replay or phish cryptographic signatures.

BlokBadge: Real MFA Without Smartphones



Split-Key Security

Badge alone is useless (needs server half).

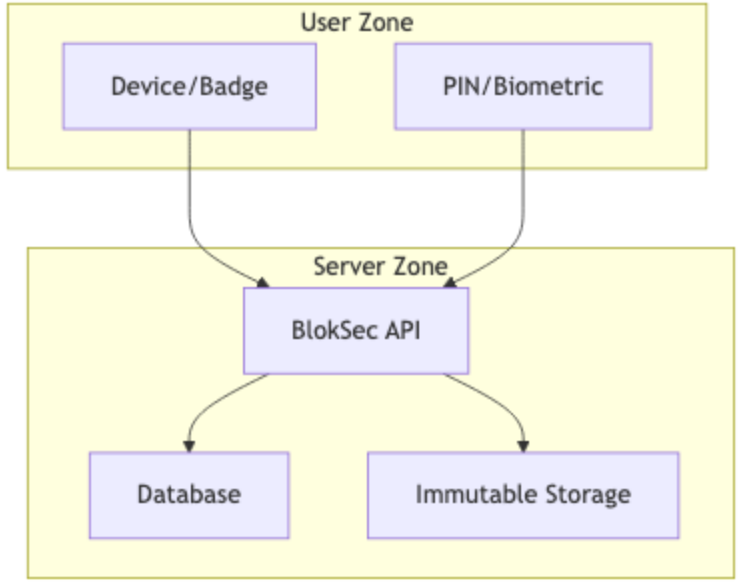
Database breach alone is useless (needs badge).

Offline cracking impossible.

Users can rotate their PIN anytime — no reprinting required.

Enterprise-Grade Security

Trust Boundaries



Immutable Storage

Prevents Key Replacement Attacks

Attackers with database access can't swap in their own public keys — records are append-only.

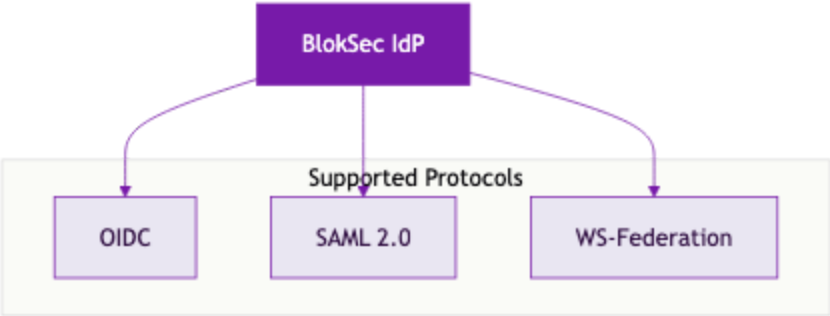
Tamper-Proof Audit Trail

Authentication events can't be altered or deleted. Regulators and forensic investigators get the truth.

vs. Traditional MFA

Threat	TOTP/SMS	BlokSec
Phishing	Vulnerable	Immune
Replay attacks	Time-window	None
Database breach	Secrets exposed	No secrets
Offline cracking	Possible	Impossible

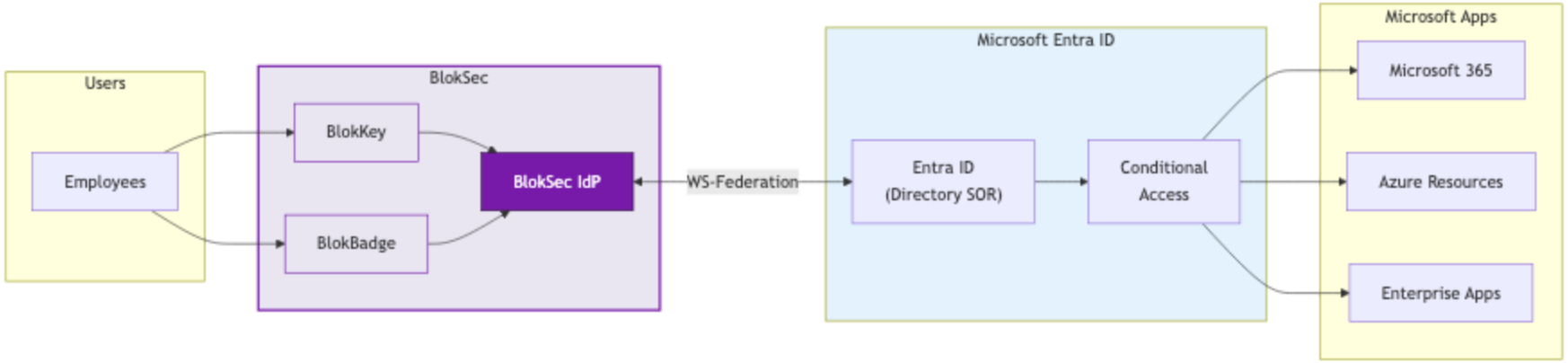
Works With Your Existing Apps



Protocol	Use Case
OIDC	Modern web/mobile apps, SPAs, APIs
SAML 2.0	Enterprise SaaS (Salesforce, Workday)
WS-Federation	Microsoft Entra ID, M365, Azure

No custom integration required. Standard protocols mean your apps already speak our language. Metadata exchange automates configuration. Deploy in hours, not months.

Integrating with Microsoft Entra ID



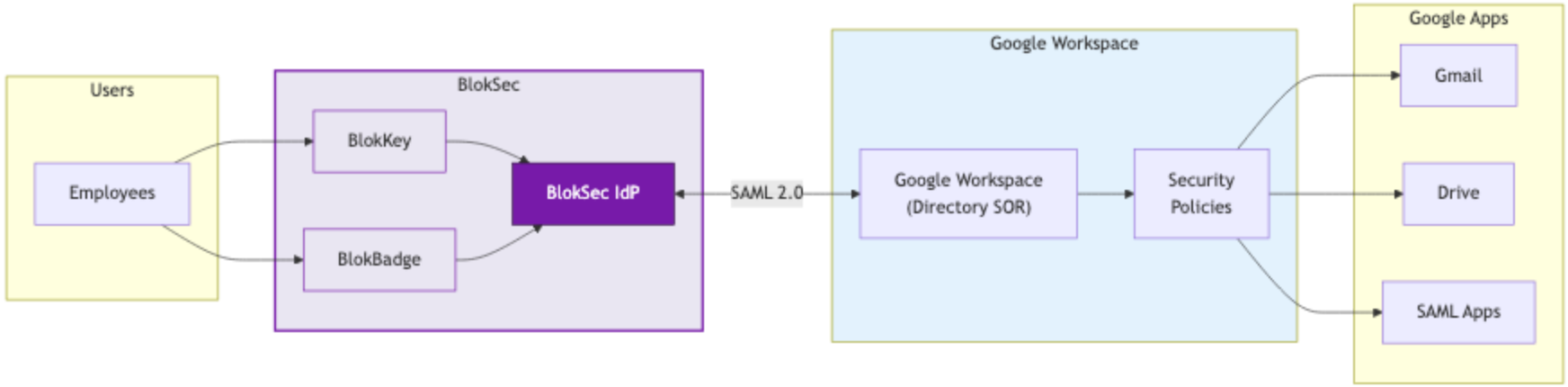
How It Works

- **Entra ID remains your directory** — Users, groups, licenses stay in Entra
- **WS-Federation** links BlokSec as external IdP
- **Conditional Access preserved** — Your policies still apply
- **Seamless SSO** to M365, Azure, and federated apps

Why It's Better

- **Eliminate Entra passwords** — Phishing-resistant auth replaces password + Authenticator
- **Frontline coverage** — BlokBadge for workers without phones (Entra can't do this)
- **Stronger MFA** — 3-factor BlokKey vs. 2-factor Authenticator
- **No Microsoft licensing changes** — Works with existing Entra tier

Integrating with Google Workspace



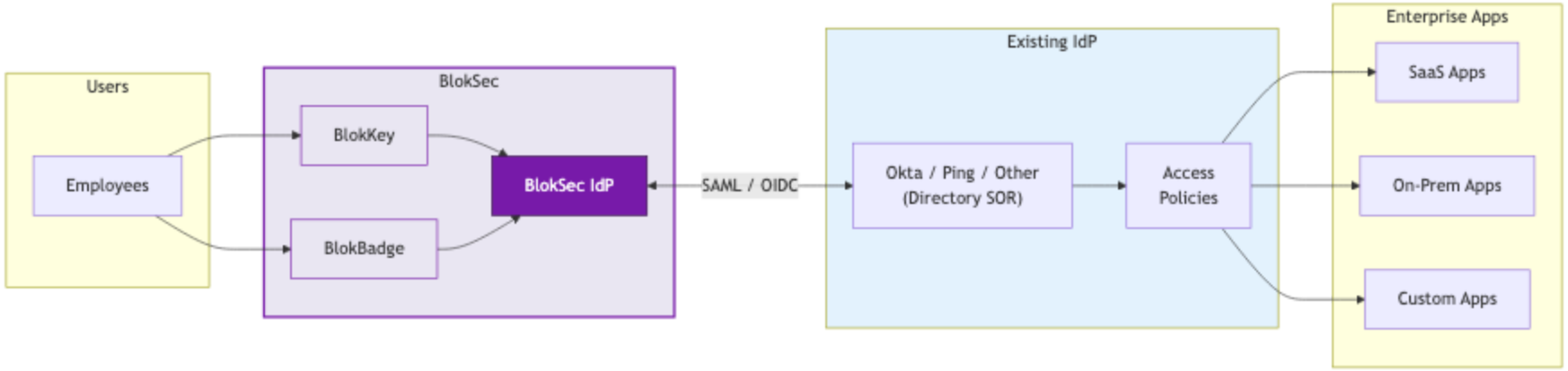
How It Works

- **Google Workspace remains your directory** — Users, groups, Drive stay in Google
- **SAML 2.0 federation** links BlokSec as third-party IdP
- **Context-Aware Access** policies still enforced
- **SSO flows** to Gmail, Drive, and SAML-federated apps

Why It's Better

- **Replace Google passwords** — No more password + 2SV prompts
- **Frontline workers covered** — BlokBadge for retail, warehouses (Google has no equivalent)
- **Phishing-resistant** — Cryptographic auth, not SMS/TOTP codes
- **Works with any Google edition** — No Workspace tier upgrade required

Integrating with Okta, Ping, and Other IdPs



How It Works

- **Your IdP remains the directory** — Users, groups, app assignments unchanged
- **Inbound federation** via SAML 2.0 or OIDC
- **Policy engine preserved** — Okta policies, Ping rules still apply
- **App integrations intact** — No changes to downstream SAML/OIDC apps

Why It's Better

- **Upgrade authentication** — Replace Okta Verify / PingID with phishing-resistant auth
- **Cover frontline workers** — BlokBadge solves the "no smartphone" gap
- **Reduce IdP MFA licensing** — BlokSec handles auth, not your IdP's MFA add-on
- **Defense in depth** — Asymmetric crypto + immutable keys on top of existing policies

Thank You

BlokSec — Passwordless authentication that works for everyone

bloksec.com